In Class Practice Decision Trees

Files: Social_Network_Ads_2.csv and new_customer_for_prediction.csv

Task: Use RapidMiner AI Studio or Python to select a decision tree and predict *Purchased* for the new customer. Choose and document your own reasonable baseline configuration.

Required work

1 **Prepare** Set Purchased as label, exclude User ID from predictors, and check types and missing values.

2 **Baseline** Build one tree with your own settings. Evaluate it by cross-validation; record folds, seed, parameters, and performance.

3 **Criteria** Compare different splitting criteria such as Information Gain and Gini Index while keeping all other settings and folds unchanged.

4 **Tune to find best tree** With cross-validation, varying tree configuration such as as depth 2–8, leaf size 1–10. Choose the best; justify your choice.

5 **Final tree** Train a new tree on all labeled records using the selected settings. Do not cross-validate this final-training step.

6 **Predict** Apply the final tree to new_customer_for_prediction.csv. Report prediction(Purchased) and both confidence values.

Evidence in the PDF

- Baseline configuration and cross-validation result
- Criterion-comparison results
- Parameter-search results showing the selected or tied best rows
- Selected final parameters and the final-tree diagram or readable rules
- Prediction and confidence values for the new record

Short answers

1. Why should User ID not be used as a predictor?
2. Why must the criterion comparison keep other settings unchanged?
3. How did you choose among tied best combinations?
4. Why is the final tree trained using all labeled records?
5. Why is its training accuracy not cross-validation performance?

Submission

Submit exactly two files:

- Coding process: .ipynb or .xml (show every steps 1-6)
- Answer document: one PDF containing the required evidence and short answers